

Programming Fundamentals – COC1071

Lab Plan – 08



Learning Objective:

- *Variables, Selection, Iterations, Functions*

- 1) Create a program that takes three angles of a triangle as input and checks if the triangle is:
 - Equilateral (all angles equal)
 - Isosceles (two angles equal)
 - Scalene (no angles equal)
- 2) Write a program that takes the current hour (in 24-hour format) and prints:
 - "Good Morning" for 5 to 11
 - "Good Afternoon" for 12 to 16
 - "Good Evening" for 17 to 20
 - "Good Night" for 21 to 4
- 3) Write a program that prints all ASCII characters along with their ASCII values from 32 to 126.
- 4) Write a program that prints numbers between 1 and 100 that are divisible by both 3 and 5.
- 5) Create a function **calculateAverage** that takes five integers and returns their average.
- 6) Write a function **updateBalance** that takes a transaction amount, adjusting the balance according to the transaction (either deposit or withdrawal). Consider that the initial balance is Rs. 1000.
- 7) Write a function **switchOnOff** that takes a boolean variable and flips its value (e.g., true to false and vice versa).
- 8) Write a function **sumOfSeries** that takes an integer n and returns the sum of the series.
- 9) Create a function **findMaxMin** that takes two integers and updates them so that the first parameter holds the maximum and the second holds the minimum.
- 10) Write a function **swapThree** that takes three integers by reference and cyclically swaps their values (e.g., a -> b -> c -> a).
- 11) Write a function **encryptChar** that takes a character by reference and shifts it by a given integer (e.g., shifting by 3 would turn 'a' into 'd').
- 12) Create a function **splitDigits** that takes an integer by reference and updates two reference variables to store the left-most and right-most digits of the integer.